

Binary RR Lyrae in the LMC from the light-travel-time effect

What we measure (the $O-C$ diagram), mock illustrations, and the pipeline

Project `rrl_lmc_binary` — working note

30 September 2026 (code: https://github.com/vasilybelokurov/rrl_lmc_binary)

Abstract

An RR Lyrae star pulsates with a period of about half a day and a very stable, sharply shaped light curve, so it can be used as a clock. If it orbits a companion, its distance from us changes periodically and the pulsation signal arrives earlier or later by the light-travel time across the orbit — minutes to tens of minutes for companions of $0.1\text{--}1\,M_\odot$ on 3–30-yr orbits. We measure this delay once per observing season for 17,492 LMC RRab stars with OGLE-II/III/IV photometry (1997–2016) and, for 6,614 of them, MACHO photometry (1992–1999). The resulting time series of delays is the $O-C$ diagram. This note explains the measurement in words and in equations, illustrates it with mock data, and documents the pipeline, its tests and its known limitations. The candidate selection is explained panel by panel in [section 5](#); results are summarized briefly in [section 6](#); the candidates are compared with the Galactic-bulge sample, and tested with Gaia photometry, in [section 7](#).

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1 What we measure, in simple words

1. **The clock.** An RRab star brightens steeply (within about an hour) and fades slowly, repeating every $P \approx 0.5\text{--}0.7$ d. The shape repeats almost exactly, so the times of the steep rise are sharp time markers.
2. **The prediction.** From the catalogue period and one reference epoch we predict when the rise should happen in any later year: the “calculated” time C .
3. **The observation.** In each observing season we find the time shift that best lines up the observed light curve with the star’s own template: the “observed” time O . The difference $O - C$ is the *delay* of that season.
4. **What changes the delay.** (i) A slightly wrong catalogue period makes $O - C$ grow linearly with time. (ii) A slow, steady change of the period (stellar evolution) bends $O - C$ into a parabola. (iii) An orbiting companion moves the RR Lyrae towards and away from us; the light has a little further or a little less distance to travel, so $O - C$ oscillates with the orbital period (figure 1). This is the light-travel-time effect (LTTE). (iv) The star itself can change its period irregularly (jumps, wander, Blazhko modulation); this also moves $O - C$ and is the main confusion.
5. **Why it works at the LMC.** The delay depends only on the size of the orbit, not on the distance to the star. A $0.6 M_\odot$ companion on a 3000-d orbit gives ± 17.5 min everywhere in the Universe. The price of the LMC distance is photometric noise: our stars are at $I \approx 19$, so one season’s delay is measured to about 4 min (median).
6. **What we look for.** A delay that oscillates smoothly and repeats, with the same amplitude in every band and survey, on top of an ordinary smooth period change; and, statistically, how many such stars there are.

2 What we measure, in equations

2.1 Light curve, ephemeris and delay

Let $T(\phi)$ be the star’s mean light curve (template) as a function of pulsation phase $\phi \in [0, 1)$, written as a Fourier series of order K (we use $K = 8$),

$$T(\phi) = \sum_{k=1}^K [a_k \cos(2\pi k\phi) + b_k \sin(2\pi k\phi)]. \quad (1)$$

With a reference period P and epoch T_0 (the OGLE-IV catalogue values), the observed magnitude is modelled as

$$m(t) = z_s + \delta m_j + \alpha_j T\left(\frac{t - \tau_j - T_0}{P}\right) + \varepsilon, \quad (2)$$

where s labels the survey segment (OGLE-II, -III, -IV, MACHO; separate zero points z_s), j the observing season, τ_j the season's delay, δm_j a mean-magnitude offset and α_j an amplitude factor (both nuisance parameters; α_j is also a Blazhko diagnostic), and ε the photometric noise. A single τ_j shifts all harmonics together: this is what a change in light-travel time does. The set $\{\tau_j\}$ versus time is the $O-C$ diagram; $\tau > 0$ means the pulsation arrives late.

2.2 Ordinary period change: the quadratic

If the period changes slowly and steadily, $P(t) = P_0 + \dot{P}t$, the n -th maximum occurs at $T_n = T'_0 + nP_0 + \frac{1}{2}\dot{P}P_0n^2$, and with $n \simeq t/P_0$

$$\tau(t) = c_0 + c_1 t + c_2 t^2, \quad c_1 = \frac{P_0 - P}{P}, \quad c_2 = \frac{\dot{P}}{2P}. \quad (3)$$

c_0 is an uninteresting constant (epoch and template-phase convention), c_1 a period error, c_2 the rate of period change. For an evolutionary $\dot{P} = 0.1 \text{ d Myr}^{-1}$ ($2.7 \times 10^{-10} \text{ d d}^{-1}$) and $P = 0.55 \text{ d}$, $c_2 = 2.5 \times 10^{-10} \text{ d}^{-1}$, i.e. $\tau \approx 2.5 \times 10^{-10} \times (4000)^2 \text{ d} = 0.004 \text{ d} \approx 350 \text{ s}$ at 4000 d from the reference epoch: comparable to the orbital signals, so the quadratic is always part of the model.

2.3 Light-travel-time effect

If the RR Lyrae (mass M_1) orbits a companion (M_2), the delay is its line-of-sight position divided by c [1]:

$$\tau_{\text{LTTE}}(t) = \frac{a_1 \sin i}{c} \frac{1 - e^2}{1 + e \cos \nu(t)} \sin(\nu(t) + \omega), \quad (4)$$

with true anomaly ν from Kepler's equation, eccentricity e , argument of periastron ω , and

$$a_1 = \left[\frac{G(M_1 + M_2)P_{\text{orb}}^2}{4\pi^2} \right]^{1/3} \frac{M_2}{M_1 + M_2}. \quad (5)$$

From the fitted P_{orb} and semi-amplitude $A \equiv a_1 \sin i / c$ one obtains the mass function

$$f(M_2) = \frac{4\pi^2 (cA)^3}{GP_{\text{orb}}^2} = \frac{(M_2 \sin i)^3}{(M_1 + M_2)^2}, \quad (6)$$

and the primary's radial-velocity semi-amplitude $K_1 = 2\pi cA / [P_{\text{orb}} \sqrt{1 - e^2}]$. The full $O-C$ model is $\tau(t) = c_0 + c_1 t + c_2 t^2 + \tau_{\text{LTTE}}(t)$.

Table 1: LTTE semi-amplitude $A = a_1/c$ for $M_1 = 0.65 M_\odot$, circular, edge-on (rrlbin.ltte.alsini_over_c).

P_{orb} [d]	$M_2 = 0.6 M_\odot$	$0.2 M_\odot$	$0.07 M_\odot$
1000	505 s	218 s	85 s
3000	1050 s (17.5 min)	453 s	177 s

2.4 Measuring one season’s delay

For fixed template (a_k, b_k, z_s) , each season’s $(\tau_j, \delta m_j, \alpha_j)$ is fitted by weighted least squares (Gauss–Newton after a coarse search over one full cycle). Its Fisher (Cramér–Rao) error is

$$\sigma_{\tau_j}^{-2} \simeq \sum_{i \in j} \frac{1}{\sigma_i^2} \left(\frac{\partial m}{\partial t} \right)_i^2 = \sum_{i \in j} \frac{\alpha_j^2 T'(\phi_i)^2}{P^2 \sigma_i^2}, \quad (7)$$

scaled by $\sqrt{\max(\chi_\nu^2, 1)}$. The steep rising branch dominates the sum. Template and delays are iterated to convergence. Because a constant delay is degenerate with the template phase, we fix the gauge so that the template’s fundamental harmonic has phase zero.

2.5 Detection statistic D

On the season delays $\{t_j, \tau_j, \sigma_j\}$ we compare two Gaussian models with variance $\sigma_j^2 + s^2$ (extra jitter s fitted in each): H_0 : [equation \(3\)](#) (+ a free MACHO offset); H_1 : $H_0 + A_s \sin(2\pi t/P_{\text{orb}}) + A_c \cos(2\pi t/P_{\text{orb}})$.

$$D = 2 \max_{P_{\text{orb}}} [\ln \mathcal{L}_1(P_{\text{orb}}) - \ln \mathcal{L}_0]. \quad (8)$$

At fixed P_{orb} and for white noise $D \sim \chi_2^2$; for N seasons of error σ and amplitude A , $D \approx NA^2/(2\sigma^2)$. D tests an orbit against a *smooth* period change plus white noise; it does not by itself reject irregular intrinsic period changes ([section 4.8](#)).

3 Mock-data illustrations

All figures in this section are synthetic (`scripts/plot_writeup_mocks.py`); the template is a typical OGLE I -band RRab template (first-harmonic amplitude 0.155 mag), and the orbit has $M_1 = 0.65$, $M_2 = 0.6 M_\odot$, $P_{\text{orb}} = 3000$ d, $e = 0.3$, $i = 90^\circ$.

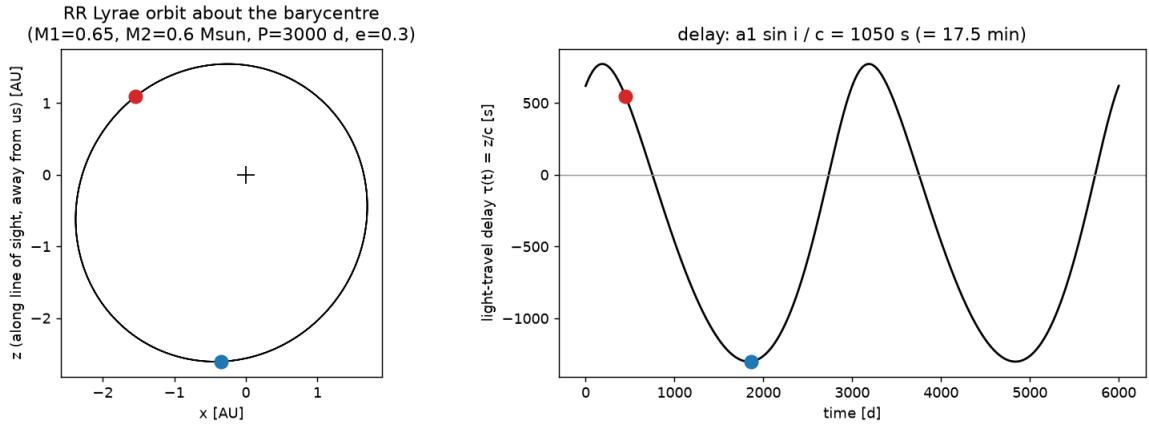


Figure 1: Left: the RR Lyrae’s orbit about the barycentre (+), with the observer below. Right: the resulting delay $\tau(t) = z/c$ ([equation \(4\)](#)); the non-sinusoidal shape comes from $e = 0.3$. Coloured dots mark the same two epochs in both panels: far side (red, late) and near side (blue, early).

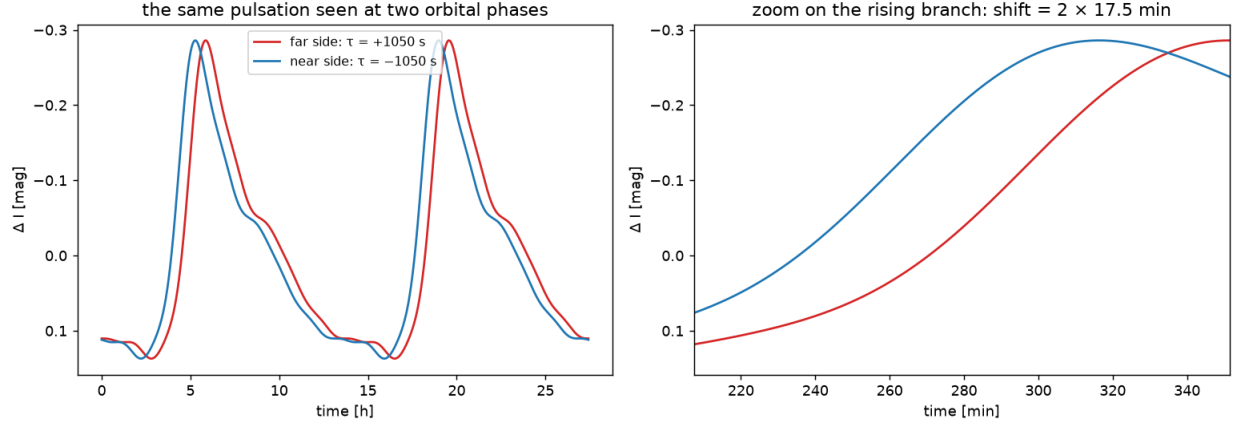


Figure 2: The same pulsation observed at the two orbital phases of [figure 1](#): the light curve is not changed in shape or amplitude, only shifted in time. Right: zoom on the rising branch, where the shift (2×17.5 min) is most easily measured.

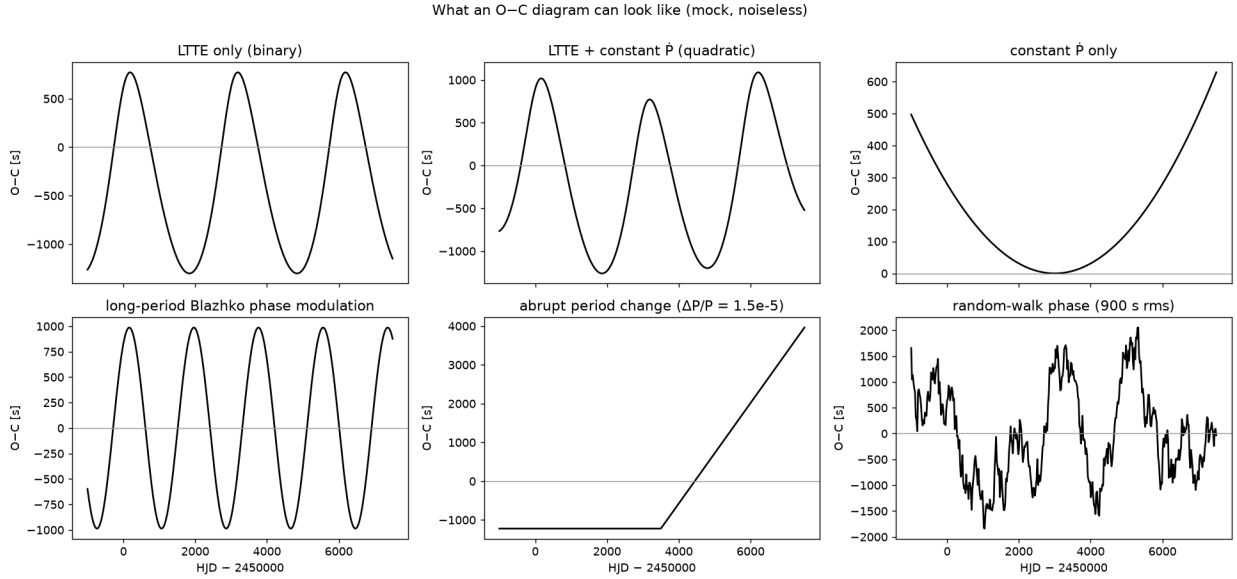


Figure 3: Noiseless mock $O-C$ diagrams over the 1992–2016 baseline. Top: binary (LTTE), binary plus constant $\dot{P}$, constant $\dot{P}$ only. Bottom: the intrinsic nuisances — long-period Blazhko phase modulation (periodic, like an orbit, but accompanied by amplitude changes), an abrupt period change (a broken line), and random-walk phase wander. Over one or two cycles all three can resemble an orbit.

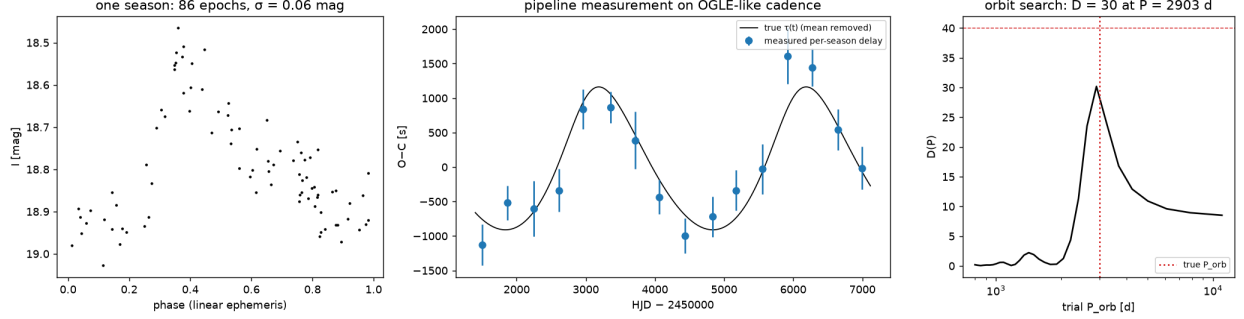


Figure 4: The pipeline on a mock star with an OGLE-III+IV-like cadence (16 seasons, 40–110 epochs each, $\sigma = 0.06$ mag). Left: one season folded on the linear ephemeris. Middle: measured season delays (median error 297 s) against the true delay. Right: the orbit search $D(P_{\text{orb}})$ (equation (8)) recovers $P = 2903$ d (true 3000 d) with $D = 30$ — a 1050-s signal is a firm but not overwhelming detection at LMC photometric precision.

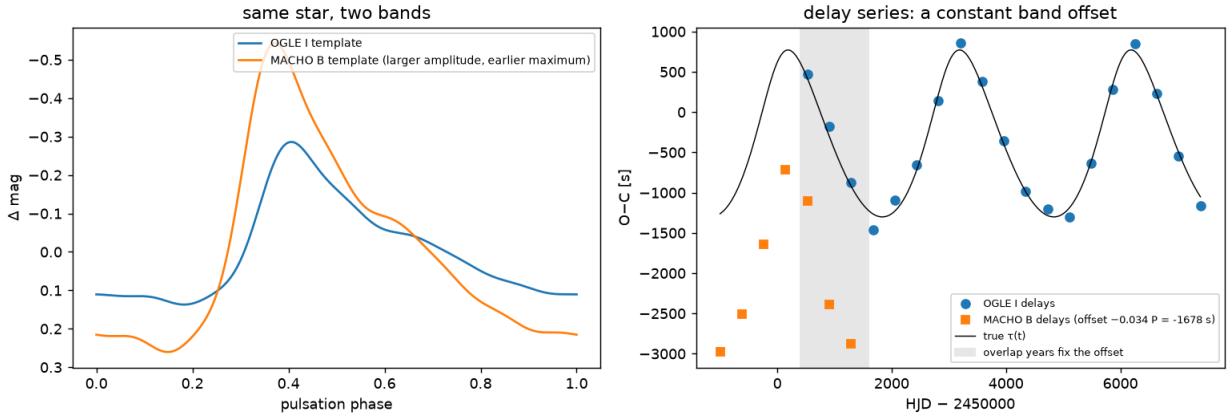


Figure 5: Two surveys in two bands. Left: MACHO B and OGLE I templates of the same star differ in amplitude and the maximum occurs 0.034 cycles earlier in B (the measured mean lag, section 4.5). Right: hence the MACHO delay series is offset by a constant from the OGLE series; the years observed by both surveys (grey) fix the offset.

4 The pipeline

4.1 Data

- **OGLE-IV** LMC RR Lyrae collection [3]: 41,471 stars; *I*-band light curves for 41,209, epochs $\text{HJD}' = 5260\text{--}8924$, where $\text{HJD}' = \text{HJD} - 2450000$ (public data mostly end at 7507, i.e. 2016). <https://www.astrouw.edu.pl/ogle/ogle4/OCVS/lmc/rrlyr/>
- **OGLE-III** [2]: 24,906 light curves, $\text{HJD}' = 2166\text{--}4540$ (2001–2008); the 2024 re-issue of the files includes **OGLE-II** epochs (from $\text{HJD}' = 447$, 1997) for 7,784 stars. Same star numbering as OGLE-IV (24,818 matches, median separation $0.05''$). <https://www.astrouw.edu.pl/ogle/ogle3/OIII-CVS/lmc/rrlyr/>
- **MACHO** *B* and *R* photometry (1992–1999) via the NCI TAP service (<https://machotap.svo.nci.org.au/ncitap/tap>), queried per tile for the RR Lyrae only (MACHO IDs from the OGLE-III cross-identification): 8,406 of 8,557 requested stars returned. We use MACHO *B*.
- **Sample for this note:** 17,492 RRab with OGLE-III and OGLE-IV light curves; 6,614 with usable MACHO data; 2,378 of these also have OGLE-II (overlap years 1997–1999).

4.2 Time system

OGLE times are HJD (UTC). MACHO `dateobs` is MJD and is converted to HJD for each star (observer at Mount Stromlo); towards the LMC, near the south ecliptic pole, the heliocentric correction is only ± 40 s (tested). Whether MACHO times refer to the start or middle of the exposure is not documented; the band-lag test in [section 4.5](#) limits any residual MACHO–OGLE clock offset to $\approx \pm 50$ s.

4.3 Seasons

A season is one observing year, with the boundary at $\text{HJD}' \bmod 365.25 = 245$ (the middle of the OGLE seasonal gap). MACHO observed the LMC nearly year-round, so gap-based seasons would merge years. Seasons with fewer than 15 epochs are dropped.

4.4 Timing fit per star and band

[equation \(2\)](#) with $K = 8$, separate zero points per survey segment, per-season $(\tau_j, \delta m_j, \alpha_j)$, 4σ clipping, template $\leftrightarrow$ delay iterations until the mask is stable and $\max |\Delta\tau| < 10^{-7}$ d, the fundamental-phase gauge, and errors from [equation \(7\)](#) scaled by $\sqrt{\max(\chi^2_\nu, 1)}$. Delays are defined modulo P and unwrapped between consecutive seasons. *Measured precision* (1000 random OGLE-III+IV RRab): median season error 253 s (10–90%: 121–510 s), strongly amplitude-dependent (476 s for $A_I < 0.4$, 113 s for $A_I > 0.8$); MACHO *B* per year: median 135 s.

4.5 Tying MACHO to OGLE

Band-dependent template shapes leave a constant offset Δ between the MACHO *B* and OGLE *I* delay series ([figure 5](#)). On 781 stars observed by both in 1997–1999: $\Delta/P = -0.034$ cycles (robust scatter 0.008), with a ~ 300 -s star-to-star scatter. The same behaviour (lag -0.035 cycles, similar period dependence and scatter) is found between OGLE’s own *V* and *I*, which share one time system: the offset is a physical band lag, not a MACHO clock error. In the *O–C* search Δ is a free parameter per star (an indicator column); for stars with OGLE-II it is fixed by the overlap years.

4.6 Common-mode timing correction

Stacking all stars’ $O-C$ residuals by year reveals a common MACHO pattern (-113 to $+40$ s) and a smaller OGLE drift ($+10$ to $+24$ s in 2006–2008). A null test (white-noise delays at the real epochs, same stacking) is consistent with zero, so these are real timing systematics; the per-(survey, year) median is subtracted from every star (median correction 12 s rms per star; the candidate list is unchanged by it).

4.7 Orbit search and the annual-sampling limit

D (equation (8)) is maximized over trial periods uniform in frequency, from 800 d to twice the baseline. Season delays sample the $O-C$ about once per year, so orbital periods below the Nyquist limit of 2 yr are aliased: in an earlier version the grid started at 300 d and a 3000-d orbit could be assigned its alias at $1/(1/365.25 - 1/3000) = 416$ d. All statistics in this note use the corrected grid. Orbits shorter than 2 yr would require within-season timing and are outside the present search.

4.8 Nuisances and discriminators

Simulated on each star’s real cadence and errors from its own fitted templates (`src/rrlbin/simulate.py`; one delay realization shared by MACHO and OGLE): constant $\dot{P}$; Blazhko (coherent phase and amplitude modulation, $P_B = 20$ –3000 d); abrupt period changes ($|\Delta P/P|$ from 10^{-6} to 2×10^{-4}); random-walk phase (50–15,000 s rms); LTTE (log-uniform $P_{\text{orb}} = 300$ – 10^4 d, $M_2 = 0.05$ – $1.5 M_\odot$, isotropic, half eccentric). Discriminators:

- amplitude constancy $\chi^2_\nu(\alpha_j) < 2$ (Blazhko veto; LTTE does not change amplitude);
- at least 1.5 orbital cycles within the baseline;
- a physical ceiling: A below a_1/c for $M_2 = 2 M_\odot$ at the fitted period (costs 4.5% of true LTTE; applied identically to simulations);
- for stars with OGLE-II: a predictive test (fit from 1997 on, predict the held-out 1992–1996 MACHO delays).

Real stars have a heavier tail of timing noise than the nuisances first simulated (jitter 90th/97th percentiles 1.6/3.8 ks), which motivated the large random-walk and large-jump classes.

4.9 Code and tests

Package `rrlbin` (`io`, `timing`, `ltte`, `simulate`, `mixture`); 20 unit tests (`pytest`): Fourier derivatives, exact recovery of noiseless delays (< 0.01 s), gauge invariance, pull distributions of noisy delays, Kepler solver and eccentric delay amplitude, LTTE amplitudes of table 1, mass function, $O-C$ search recovery and null calibration, the annual-Nyquist grid, predictive score, MJD→HJD. Long runs use a chunked, resumable runner. Full log: `JOURNAL.md`.

5 Selection of candidates

Figure 6 summarizes the $O-C$ fits of all 17,492 RRab (after the common-mode correction and on the Nyquist-safe period grid) together with 23,552 simulations run through the same pipeline on the real cadences of 1,472 MACHO+OGLE stars (`results/inject/macho_v2.parquet`; 2 null, 2 Blazhko, 1 small jump, 1 small random walk, 2 large jumps, 2 large random walks and 4 LTTE per star). All numbers below are from `scripts/selection_numbers.py` (`results/real/selection_numbers.json`).

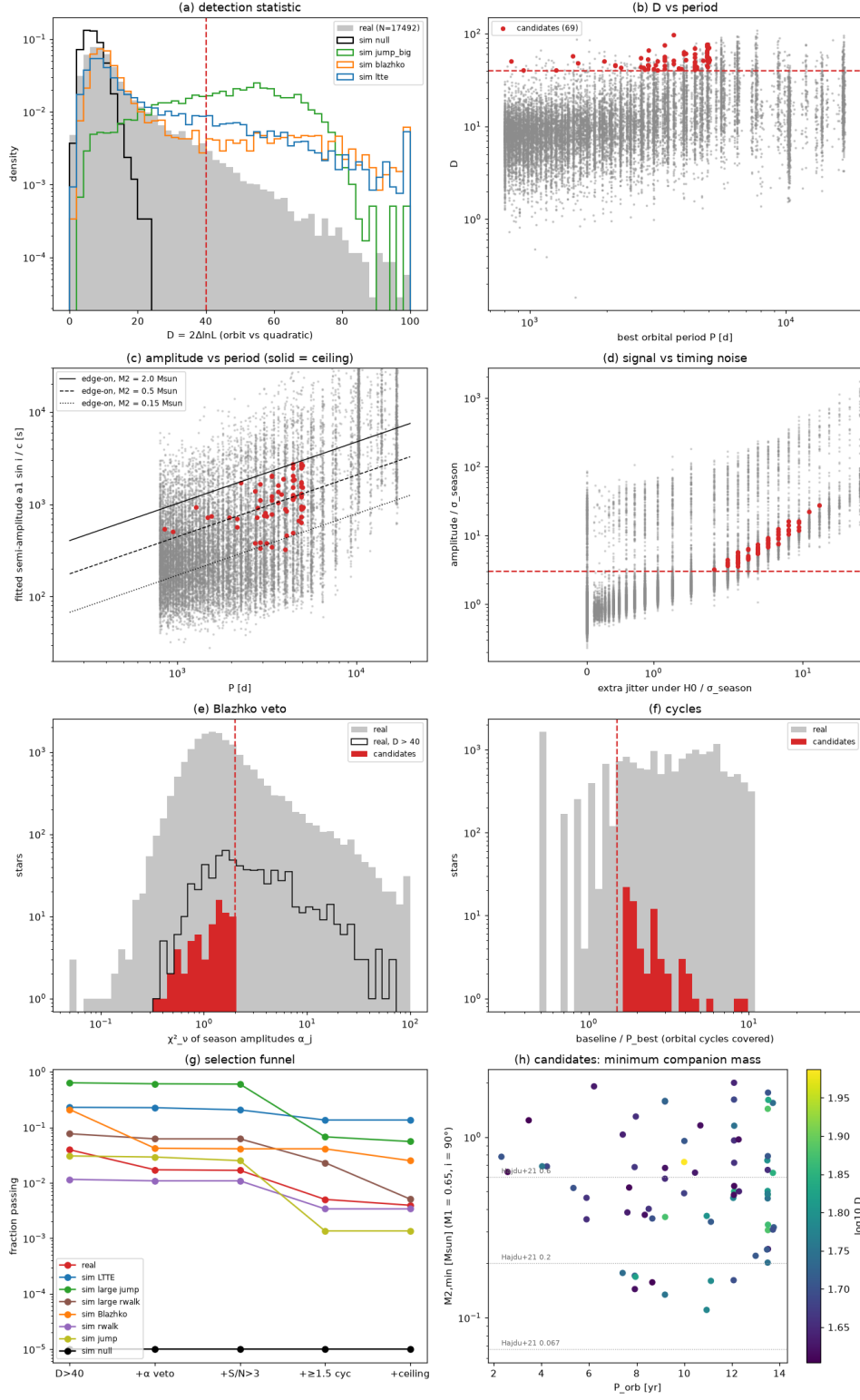


Figure 6: Summary statistics of the $O-C$ fits and the candidate selection. Grey: all real RRAb; red: the 69 candidates; coloured lines: simulated classes. Panels are described in [section 5.1](#); the cuts in [table 2](#).

5.1 The panels

(a) Detection statistic.

Distribution of D (equation (8)) for real stars (grey) and for four simulated classes. White-noise nulls (black; only a constant $\dot{P}$ and the photometric noise) never exceed $D = 23$ in 2,944 simulations; their per-star false-alarm thresholds are $D = 12.4$ (5%), 15.6 (1%) and 20.2 (0.1%). Real stars have a long tail extending far beyond the nulls: at $D > 40$ they number 691 (4.0%) where white noise predicts none. Simulated orbits (blue), Blazhko modulation (orange) and large abrupt period changes (green) all populate that tail, so D alone cannot separate orbits from intrinsic period changes.

(b) D versus best period.

Each star's D at the period that maximizes it. The search starts at 800 d (section 4.7); the vertical banding is the discreteness of the frequency grid (nearly all MACHO stars share an 8,350-d baseline and hence the same grid points). Candidates (red; 81% of them) concentrate at 2,500–5,000 d, the range in which a 24-yr baseline covers about 2–4 cycles and the search is most sensitive.

(c) Amplitude versus period.

Fitted semi-amplitude $A = a_1 \sin i / c$ with the edge-on LTTE amplitude for $M_2 = 0.15, 0.5$ and $2 M_\odot$ ($M_1 = 0.65 M_\odot$). The solid line is the physical ceiling used in cut 5. The large cloud above it at long periods consists of stars whose $O-C$ changes by a large fraction of a pulsation cycle — intrinsic period changes far larger than any companion of $\leq 2 M_\odot$ can produce. Candidates lie between the 0.15 and $2 M_\odot$ curves, about half (52%) above the $0.5 M_\odot$ curve.

(d) Signal versus timing noise.

Orbit amplitude and extra jitter (fitted under H_0), both in units of the median season error. Stars with no extra jitter under H_0 form the column at zero. A real orbit that is not modelled under H_0 is absorbed as jitter, so strong orbits move up and to the right together; the discrete columns are the jitter grid. Cut 3 (dashed) removes signals that are not resolved season by season.

(e) Blazhko veto.

χ_ν^2 of the per-season amplitude factors α_j about their mean. A light-travel delay shifts the light curve without changing its amplitude ($\chi_\nu^2 \approx 1$), whereas Blazhko modulation changes it: the median χ_ν^2 is 0.97 for simulated orbits and 0.97 for nulls, but 1.8 for simulated Blazhko stars and 1.4 for real stars. The black histogram (real, $D > 40$) is shifted to large χ_ν^2 : most high- D real stars change amplitude from season to season.

(f) Cycles covered.

Baseline divided by the best period. Real stars with long best periods (< 1.5 cycles) are numerous; at fewer than 1.5 cycles a sinusoid is not distinguishable from a smooth curve (cubic trend, broken line, random walk).

(g) Selection funnel.

Fraction of each class passing the cuts applied in sequence. Real stars (red) fall from 4.0% to 0.39%; simulated orbits (broad prior, blue) from 23% to 13.6%; the most orbit-like nuisance, large abrupt period changes (green), from 64% to 5.6%; white-noise nulls pass none.

(h) Candidates.

Minimum companion mass (edge-on, $M_1 = 0.65 M_\odot$, from equation (6)) against orbital period, coloured by $\log D$; dotted lines mark the three companion-mass groups suggested for the Galactic

bulge by Hajdu et al. [5]. This panel is descriptive: which of these are binaries is not established (section 6).

5.2 The cuts and why

Table 2: Candidate cuts, applied in order. Columns: surviving real stars (all / with MACHO), and the fraction of each simulated class still passing after the cut (LTTE: broad prior, $P_{\text{orb}} = 300\text{--}10^4$ d, $M_2 = 0.05\text{--}1.5 M_\odot$).

	cut	real	LTTE	Blazhko	jump	large jump	large r.w.	null
1	$D > 40$	691 / 496	0.230	0.210	0.031	0.638	0.077	0
2	$\chi^2_\nu(\alpha) < 2$	301 / 232	0.227	0.042	0.029	0.611	0.062	0
3	$A/\sigma_{\text{season}} > 3$	294 / 225	0.207	0.041	0.025	0.604	0.062	0
4	≥ 1.5 cycles	88 / 71	0.136	0.041	0.001	0.068	0.023	0
5	$A < a_1/c (M_2 = 2 M_\odot)$	69 / 62	0.136	0.025	0.001	0.056	0.005	0

1. $D > 40$. With 17,492 stars, fewer than one white-noise false alarm requires a per-star false-alarm probability below $\sim 6 \times 10^{-5}$, i.e. well beyond the largest null value ($D = 23$ in 2,944 simulations). $D > 40$ is chosen conservatively high because the real contaminants are not white noise; lowering it mainly admits more intrinsic period changes (panel a).
2. **Amplitude constancy**, $\chi^2_\nu(\alpha) < 2$. Motivated by the physics: the LTTE is a pure time shift (figure 2), so any season-to-season amplitude change points to Blazhko modulation (or changing blends). The cut removes 80% of simulated Blazhko stars that passed cut 1 ($0.210 \rightarrow 0.042$) while keeping 99% of the orbits ($0.230 \rightarrow 0.227$), and 56% of real stars with $D > 40$.
3. $A/\sigma_{\text{season}} > 3$. The orbital amplitude must be large compared with a single season’s error, so that the curve is seen season by season and is not only an integrated excess; this mainly guards against low-amplitude, high-jitter solutions and has little effect after cuts 1–2.
4. **At least 1.5 orbital cycles**. The decisive cut against intrinsic period changes: a sinusoid fitted over less than about one cycle mimics a smooth trend. It removes 89% of the large abrupt period changes ($0.604 \rightarrow 0.068$) and 63% of the large random walks, at the cost of a third of the simulated orbits — those with periods longer than about 2/3 of the baseline, which cannot be confirmed as periodic anyway.
5. **Physical ceiling**. The fitted amplitude must be achievable with $M_2 \leq 2 M_\odot$ at the fitted period (panel c). At this stage it removes no simulated orbits ($0.136 \rightarrow 0.136$) but most of the remaining large random walks ($0.023 \rightarrow 0.005$) and 19 real stars. Over the full sample 4.5% of true orbits exceed the ceiling because of noise; the cost is measured, and the cut is applied identically to data and simulations.

The cuts were set on physical grounds and on the simulations, not tuned on the real candidates. They are not a clean binary selection: large abrupt period changes still pass at 5.6% and Blazhko at 2.5% (table 2), and their true incidence among LMC RRab is unknown.

6 Status

- **Candidates:** 69 stars (62 with MACHO), 65 of which were also selected before the period-grid fix. Several show smooth $O-C$ curves over two or more cycles (figure 7); periods 2.3–13.8 yr (median 10.9 yr); implied $M_{2,\min} = 0.11\text{--}2.0 M_{\odot}$ and $K_1 = 1.8\text{--}16.5 \text{ km s}^{-1}$.
- **Completeness** of the cuts for isotropic orbits (half eccentric), MACHO+OGLE cadence: $M_2 = 0.4\text{--}1.5 M_{\odot}$: 46% ($P_{\text{orb}} = 0.8\text{--}3 \text{ kd}$), 34% (3–10 kd); $M_2 = 0.15\text{--}0.4 M_{\odot}$: 7% and 12%; $M_2 < 0.15 M_{\odot}$: $\leq 1\%$.
- **Upper limit** (all 62 MACHO candidates treated as binaries; 95% Poisson limit 76.6 of 6,614 stars): $f < 2.8\%$ for $M_2 = 0.4\text{--}1.5 M_{\odot}$ and $f < 11\%$ for $M_2 = 0.15\text{--}0.4 M_{\odot}$, both at $P_{\text{orb}} = 1\text{--}10 \text{ kd}$.
- **Not yet a measurement:** the population mixture fit of the binary fraction is only weakly identified (it depends on the assumed nuisance shapes by a factor $\gtrsim 2$). The main contaminant is large abrupt period changes.

#2 OGLE-LMC-RRLYR-08101 $P_{\text{puls}} = 0.51526 \text{ d}$, $I = 18.809$, $A_I = 0.823$ | $P_{\text{orb}} = 4419 \text{ d}$ (12.1 yr), $a_1 \sin i/c = 2050 \text{ s}$, $f(M) = 0.473 M_{\text{sun}}$, $M_{2,\min} = 1.16 M_{\text{sun}}$, $K_1 = 10.1 \text{ km/s}$ | $D = 56$, pred. score = n/a

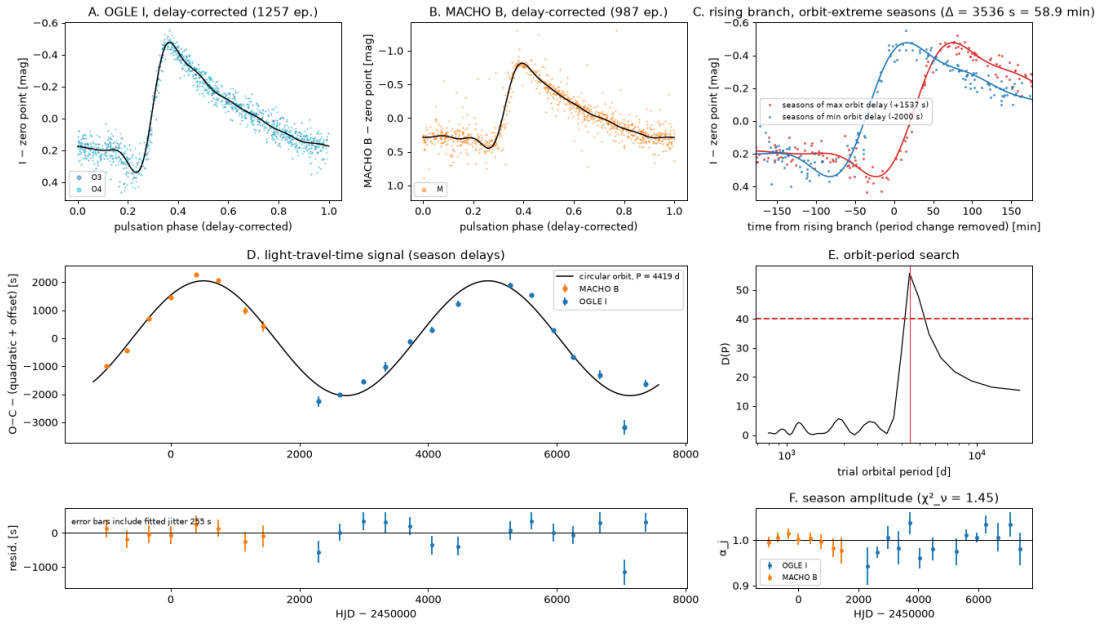


Figure 7: Example candidate sheet. A, B: delay-corrected folds (OGLE I , MACHO B) with templates. C: raw OGLE points on the rising branch in the seasons of maximum and minimum orbital delay, with the period change removed — the whole waveform is shifted by about an hour. D: $O-C$ with the circular-orbit fit and residuals (error bars include the fitted jitter). E: $D(P_{\text{orb}})$. F: season amplitudes.

7 Part C: the candidates compared with the Galactic-bulge sample

This section uses the v3 pipeline (Part A of the refit: MACHO R , harmonic coherence, whole-cycle band alignment, band-lag priors, smooth-wander noise model). Selection thresholds and

contamination rates are **provisional** until the v3 simulations (Part B) are complete. Scripts: `scripts/partC_*.py`, `compare_hajdu.py`, `gaia_timing*.py`.

7.1 Our v3 candidates in brief

The cuts of [table 2](#) applied to the v3 statistics select 75 stars (69 with MACHO). Each was fitted with a full Keplerian LTTE orbit (Irwin delay + quadratic + MACHO offsets with priors + white jitter; multi-start least squares, residual bootstrap; recovery of injected $e = 0, 0.4, 0.7$ orbits verified). Independent of the detection statistic we examined: the per-season amplitude constancy and harmonic coherence in every band, the out-of-sample prediction of the 1992–1996 MACHO seasons, crowding indicators from Gaia DR3 (MACHO candidates are indistinguishable from their parent population; KS $p = 0.2$ – 0.99), and the Keplerian fit quality. Grouping:

- **Tier 1** (11): Keplerian fit not at the eccentricity bound and covering ≥ 1.5 cycles, $\chi_\nu^2 < 1.5$; amplitude and coherence $\chi_\nu^2 < 1.5$; no crowding flag; predictive score > 2 . All have MACHO B , R and OGLE I . $P_{\text{orb}} = 5.4$ – 14.5 yr, $e = 0.15$ – 0.57 , $M_{2,\text{min}} = 0.14$ – $0.96 M_\odot$, $K_1 = 2.5$ – 11.5 km s^{-1} , $V = 19.2$ – 19.7 .
- **Tier 2** (17): at most one soft flag; predictive score > 0 or unavailable. **Tier 3** (47): the rest.

7.2 The bulge sample of Hajdu et al. (2021)

Hajdu et al. [5] examined $\sim 27,500$ OGLE-III/IV O–C diagrams of bulge RRab (data to 2019; mean/median baselines 10.5/7.4 yr), pre-selected ~ 400 by eye, and retained 87 after MCMC LTTE fits including a period-change term, in three quality classes (Q1: 25, P_{orb} uncertainty $< 1\%$; Q2: 32; Q3: 30, problematic fits). Their Table 1 was parsed from the arXiv source (`scripts/parse_hajdu2021.py`).

Follow-up status. We found no published confirmation or refutation of any of the 87. A 2026 study co-authored by Hajdu and Prudil states that “only a single RRL is confirmed as belonging to a binary system” [9], namely TU UMa, a field star [7]. Radial-velocity tests of Galactic field timing candidates have been largely negative (15 of 19 without binarity, Barnes et al. 10; KIC 2831097 ruled out, Poretti et al. 11). The status of the bulge candidates is therefore *untested*, not refuted. (Literature search: `docs/reviews/hajdu_followup.md`; an ADS full-text query could not be run.)

7.3 Side-by-side comparison

Table 3: Medians per class. $A = a_1 \sin i/c$; σ_A/A , σ_P/P , σ_e : Hajdu posterior standard deviations, ours bootstrap half 16–84% widths; cycles = baseline/ P_{orb} (Hajdu baseline *assumed* 6700 d, 2001.4–2019.7).

class	N	P_{orb} [d]	A [s]	e	$M_{2,\text{min}}$	K_1	σ_A/A	σ_P/P	σ_e	cycles
Hajdu Q1	25	3437	779	0.27	0.44	6.5	0.017	0.005	0.024	1.9
Hajdu Q2	32	3477	355	0.31	0.14	2.6	0.037	0.015	0.054	1.9
Hajdu Q3	30	4910	534	0.35	0.20	3.4	0.092	0.053	0.065	1.4
LMC Tier 1	11	2914	804	0.34	0.49	6.2	0.067	0.011	0.102	2.9
LMC Tier 2	17	3131	736	0.32	0.44	6.2	0.070	0.016	0.127	2.4

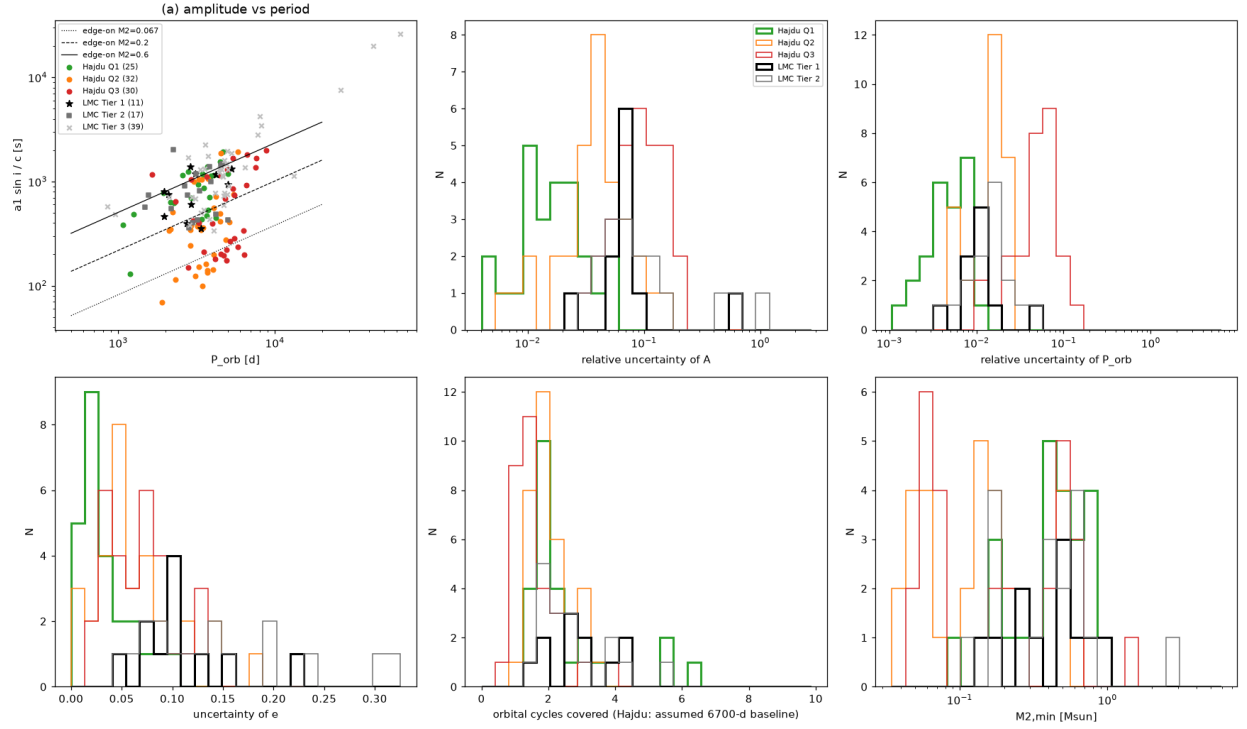


Figure 8: Our candidates (Tier 1: stars; Tier 2: squares; Tier 3: crosses) against the Hajdu Q1/Q2/Q3 bulge candidates: (a) LTTE amplitude vs orbital period with edge-on curves for $M_2 = 0.067, 0.2$ and $0.6 M_\odot$ (the three groups suggested by Hajdu et al.); (b, c) relative uncertainties of A and P_{orb} ; (d) uncertainty of e ; (e) orbital cycles covered; (f) minimum companion mass.

Where the bulge sample is stronger: precision. At $I \approx 15\text{--}16$ the bulge stars are 3–4 mag brighter; their orbital parameters are 2–4 times more precise (σ_A/A 1.7% vs 6.7%, σ_P/P 0.5% vs 1.1%, σ_e 0.02 vs 0.10 for Q1 vs Tier 1). Their seasons were also split into segments (more O–C points). Their posterior widths assume white timing noise beyond a constant $\dot{P}$, which understates the uncertainty if the period wanders (section 4.8).

Where our sample is stronger: robustness against the dominant false-positive mechanism. The main way an RR Lyrae imitates an orbit is slow intrinsic phase wander (irregular period changes are common in every population studied; e.g. Alcock et al. 12 for LMC RRc). Four properties of our Tier 1 address it directly:

1. **Independent instruments.** The orbit must be the same in MACHO (B, R ; Mount Stromlo, 1992–1999) and OGLE (I ; Las Campanas, 1997–2016). The bulge O–C diagrams come from OGLE alone.
2. **More cycles:** median 2.9 vs 1.9. A single smooth excursion, the commonest form of wander, cannot produce two full, matching cycles.
3. **An out-of-sample prediction passed:** every Tier 1 orbit, fitted to the 1997–2016 data, predicts the 1992–1996 MACHO delays better than the no-orbit model (score > 2). The bulge analysis has no equivalent (except one extrapolation check).
4. **Calibrated contamination:** thresholds and nuisance rates come from simulations on the real cadences with a noise model matched to the real stars (smooth wander, section 4.8); the bulge pre-selection was by eye, without a false-alarm rate.

Where they are alike. Our sensitivity limit ($A \gtrsim 350$ s) restricts us to the high-amplitude regime of the bulge Q1 class: medians $A \approx 800$ s, $M_{2,\min} \approx 0.45\text{--}0.5 M_\odot$, $K_1 \approx 6 \text{ km s}^{-1}$ in both (figure 8a, f). We cannot test the low-mass Q2/Q3 population ($M_{2,\min} \sim 0.1\text{--}0.2 M_\odot$).

Expectation. The strongest Tier 1 candidates have passed exactly the tests that phase wander tends to fail, so we expect them to fare better than a typical bulge candidate. How many could still be wander will be quantified by the empirical-noise class of Part B (pass rate of the full Tier 1 criteria). Confirmation is, however, harder: at $V \approx 19.4$, measuring $K_1 \sim 6 \text{ km s}^{-1}$ requires $\sim 1\text{--}2 \text{ km s}^{-1}$ per epoch after removing a $\sim 60 \text{ km s}^{-1}$ pulsation curve (published LMC RR Lyrae RVs reach $10\text{--}35 \text{ km s}^{-1}$ per epoch).

7.4 A photometric test with Gaia

Gaia DR3 epoch G photometry (2014.6–2017.4) exists for 62 of the 75 candidates (median 34 usable transits, per-transit S/N ≈ 59). We fitted a G -band template and two delays, one per half of the DR3 window; the unknown $G\text{--}I$ band lag cancels in their difference $\Delta\tau$, which is compared with each Keplerian prediction.

- **Calibration on 228 quiet non-candidate stars** (Gaia $\Delta\tau$ vs OGLE’s own $\Delta\tau$ at the same epochs, OGLE errors included): robust scatter of $z = 1.64$ (2.2% beyond 5σ) $\Rightarrow$ Gaia errors inflated by 1.64.
- **DR3 result:** only 4 of 53 testable candidates are informative ($|\Delta\tau_{\text{pred}}| > 2\sigma$). Two agree (16755, Tier 1, $z = 0.8$; 13469, $z = 0.3$); two disagree (13392, Tier 2, $z = -6.7$; 01106, Tier 3, $z = +3.5$), versus ~ 0.5 expected by chance from the control tail. DR3 is too sparse for most orbits.

- **DR4 forecast** (expected December 2026; 2014.6–2020.1, *assumed* $\sim 2\times$ the DR3 epochs; bins 2014.6–2017.4 and 2017.4–2020.1, the latter entirely after the public OGLE-IV data): testable at $> 2\sigma$ for 6 of 9 Tier 1 (5 at $> 3\sigma$), 8 of 13 Tier 2, 13 of 31 Tier 3. DR4 can therefore confirm or reject most of the strongest candidates photometrically, at no observing cost. OGLE-IV photometry after 2016 (not public) would be a stronger test still.

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